

SPRINGBANK, AB, Canada

WMO: 718600

Lat: 51.100N

Lon: 114.370W

Elev: 1201

StdP: 87.71

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.5	(c) -26.9	(d) -35.0	(e) 0.2	(f) -30.3	(g) -31.1	(h) 0.2	(i) -26.9	(j) 13.5	(k) 4.7	(l) 12.2	(m) 3.8	(n) 2.5	(o) 320	(p) 0.719

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.3	(c) 27.5	(d) 15.6	(e) 25.6	(f) 14.9	(g) 23.8	(h) 14.4	(i) 17.1	(j) 24.7	(k) 16.1	(l) 23.5	(m) 15.1	(n) 22.4	(o) 4.2	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.2	11.7	20.3	13.1	10.9	19.0	12.1	10.2	17.7	52.6	24.5	49.4	23.6	46.4	22.6	20.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	9.6	8.3	-34.7	30.5	4.7	2.2	-38.1	32.0	-40.8	33.3	-43.5	34.5	-46.9	36.1
			-34.8	18.9	4.7	1.0	-38.2	19.6	-40.9	20.2	-43.5	20.7	-46.9	21.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-8.8	-7.8	-3.4	2.9	8.1	12.0	15.0	14.5	9.6	3.8	-3.6	-8.4
	DBStd	10.52	8.63	8.39	7.91	4.92	4.46	3.13	2.79	2.99	4.33	4.49	7.25	8.14
	HDD10.0	3056	582	498	417	216	90	16	2	3	57	197	409	569
	HDD18.3	5650	840	731	675	463	318	192	108	123	262	451	659	828
	CDD10.0	454	0	0	0	2	30	74	157	141	45	4	1	0
	CDD18.3	7	0	0	0	0	0	1	4	2	0	0	0	0
	CDH23.3	484	0	0	0	2	22	40	179	201	40	0	0	0
	CDH26.7	88	0	0	0	0	1	6	32	44	6	0	0	0
Wind	WSAvg	3.5	3.4	3.4	3.5	3.8	3.9	3.7	3.3	3.1	3.2	3.7	3.2	3.6
Precipitation	PrecAvg	498	15	12	23	35	75	113	66	60	44	22	18	16
	PrecMax	686	27	25	58	84	135	350	159	124	108	53	55	48
	PrecMin	329	2	0	7	13	25	44	20	13	8	5	1	3
	PrecStd	83	7	7	13	19	33	61	35	30	25	12	12	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	11.2	16.0	21.6	25.9	27.3	29.7	30.2	26.9	20.9	14.8	10.5
		MCWB	3.5	4.2	6.2	9.6	12.8	15.9	16.6	16.1	13.2	9.8	6.8	3.0
	2%	DB	6.9	7.9	12.8	17.9	23.3	24.1	27.2	27.8	24.2	17.4	11.1	6.8
		MCWB	2.0	2.4	5.1	7.7	11.7	14.1	16.4	15.3	13.2	8.9	5.0	1.4
	5%	DB	5.3	5.6	9.9	15.0	20.7	22.0	25.4	25.6	21.7	14.6	8.4	4.6
		MCWB	1.1	1.0	3.7	6.4	10.5	13.1	15.7	15.1	12.4	7.2	3.2	0.3
	10%	DB	3.1	3.4	7.0	12.1	18.0	20.1	23.4	23.4	19.1	12.2	5.9	2.4
		MCWB	-0.5	-0.5	1.9	4.7	9.5	12.2	15.1	14.8	11.3	5.9	1.8	-1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	4.6	6.9	9.9	13.4	16.5	18.7	18.1	15.2	10.9	7.2	3.5
		MCDB	8.5	10.2	14.1	21.1	23.3	24.9	25.5	26.0	24.5	19.0	13.2	8.7
	2%	WB	2.4	2.6	5.5	8.3	12.3	15.0	17.5	16.8	13.6	9.2	5.3	1.8
		MCDB	6.6	7.5	12.4	17.0	21.5	22.0	24.8	24.4	22.3	16.3	10.5	6.4
	5%	WB	1.1	1.3	3.7	6.7	11.2	13.9	16.5	15.9	12.6	7.6	3.5	0.3
		MCDB	5.2	5.3	9.4	14.5	19.4	20.7	23.9	23.3	20.6	14.1	8.0	4.1
	10%	WB	-0.4	-0.3	2.0	5.1	10.0	12.7	15.6	14.9	11.5	6.1	1.8	-1.2
		MCDB	3.0	3.0	6.9	11.3	17.2	19.0	22.6	22.6	18.4	11.8	5.7	2.2
Mean Daily Temperature Range	5% DB	MDBR	11.3	11.9	12.4	12.9	13.5	12.9	14.3	14.9	14.0	13.2	11.3	10.9
		MCDBR	13.2	13.5	16.0	18.5	19.1	17.3	18.5	20.1	20.1	17.1	14.7	12.6
		MCWBR	9.6	9.1	9.7	10.2	9.3	8.7	9.1	9.6	10.4	9.9	9.2	8.6
	5% WB	MCDBR	12.8	13.4	15.7	17.8	17.8	15.5	16.4	17.3	18.3	16.7	14.4	12.6
		MCWBR	9.6	9.3	9.7	10.1	8.9	8.2	8.7	9.0	9.9	9.9	9.3	8.9
Clear-Sky Solar Irradiance	taub		0.249	0.267	0.279	0.334	0.340	0.346	0.340	0.355	0.310	0.277	0.252	0.236
	taud		2.443	2.446	2.478	2.326	2.376	2.395	2.416	2.387	2.485	2.562	2.510	2.426
	Ebn at Noon		810	884	936	907	914	909	908	875	887	866	803	769
	Edh at Noon		56	74	87	114	114	113	109	106	86	65	52	48
All-Sky Solar Radiation	RadAvg		1.12	2.08	3.30	4.58	5.36	5.67	6.33	5.23	3.69	2.28	1.22	0.88
	RadStd		0.07	0.15	0.23	0.32	0.44	0.45	0.43	0.37	0.41	0.23	0.08	0.06

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page